

KATHERINE MARINE

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EDUCATION

Dartmouth College, Hanover, NH

June 2028

Bachelor of Arts, intended Major in Quantitative Social Science; Minor in Economics

GPA 3.90/4.0

Citations for Academic Excellence: *Math 50, Cognitive Science 01*

The Pingry School, Martinsville, NJ

June 2024

Honor Roll, all semesters; *summa cum laude*

RELEVANT EXPERIENCE

Dartmouth Tech Consulting Group, Hanover, NH

January 2025 – Present

Executive Developer and Treasurer

- Provide pro-bono tech consulting services in software engineering and data science to early-stage startups across the world
- Partnered with an independent basketball trainer to scope and build a customer-facing booking platform: designed backend database and developed front end in HTML/JavaScript to support client booking and lead capture
- Manage budgeting and resource allocation for the org as Treasurer, supporting four concurrent client engagements

My Cancer Family, New York, NY

May 2025 – September 2025

Data Science and Product Intern

- Conducted exploratory data analysis on user engagement data for a nonprofit supporting families affected by cancer to identify highest-retention content types, delivering recommendations that helped the product team shift their marketing strategy and start publishing their newsletter once a week, resulting in 35% more repeat visitors over the next quarter.
- Built SQL pipeline queries to clean and join 12 months of engagement datasets from multiple sources, consolidating metrics into a single reporting view that replaced manual data pulls and allowed for more efficient performance tracking.
- Co-led rebuild of the organization's web and mobile platforms, migrating from WordPress to a more scalable infrastructure, optimizing load time and enabling cleaner downstream analytics.

Dartmouth Evergreen AI, Hanover, NH

May 2025 – Present

AI Product and Content Developer

- Joined a Dartmouth-sponsored student mental health initiative as one of the first hires, contributing across engineering and content as the project scaled from early prototype to functional product.
- Develop conversation logic trees in Dialogflow to build a model agent that guides students through supportive, rule-based conversations with intent recognition and rooted in behavioral science.
- Designed and analyzed A/B tests on app and dialogue variations, using engagement data and user response to inform decisions on content and product direction.

Dartmouth Athletics, Hanover, NH

January 2026 – Present

Data Analytics Intern

- Analyze athlete tracking data from Catapult wearable devices across 2 Division I teams to evaluate performance metrics and inform coaching decision-making.
- Build R-based statistical models comparing GPS and inertial movement (IMA) acceleration data, identifying which indoor metrics are reliable substitutes for GPS and delivering recommendations for the department's tracking strategy.
- Present findings and visual reports to coaches highlighting trends in athlete workload and performance, and deliver actionable insights to minimize injury and optimize athlete preparedness.

Dartmouth Sports Analytics Club, Hanover, NH

December 2025 – Present

Vice President

- Selected for Syracuse University Football Analytics Blitz; led quantitative analysis of NFL two-high safety coverage trends and delivered a data-driven offensive game plan with specific route concepts tailored to the Buffalo Bills.
- Mentor 30+ club members in data cleaning, modeling, and visualization techniques using Python/R for real-world sports data.
- Oversee student-led projects spanning machine learning, network analysis, and statistical inference.

Dartmouth Women in Business, Hanover, NH

October 2024 – Present

Active Member

- Participate in panels and case competitions on business analytics, financial modeling, and strategic decision-making.
- Worked with peers to analyze business case studies, identify key performance indicators, and develop actionable insights for marketing and operational challenges, culminating in a pitch-style presentation.

SKILLS & INTERESTS

Technical: Python (Pytorch, Pandas, Numpy), R, SQL, Tableau, Java, NLP, Git, Claude Code, Scikit-learn, Microsoft Office, Excel

Interests: Sports (Premier League, NFL, NBA), Artificial Intelligence, Marathon running, Triathlon team, Lord of the Rings trilogy